University of St.Gallen

Course and Examination Fact Sheet: Spring Semester 2026

8,015: Sustainable Innovation through Human-centered Design

ECTS credits: 3

Overview examination/s

(binding regulations see below)
decentral - Presentation, Analog, Group work group grade (30%)
Examination time: Term time
decentral - Active participation, Analog, Individual work individual grade (10%)
Examination time: Term time
decentral - Written work, Digital, Individual work individual grade (60%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
8,015,1.00 Sustainable Innovation through Human-centered Design -- English -- Uebernickel Falk

Course information

Prerequisites

This course is open to students from a wide range of academic and professional backgrounds. No specific prior coursework is required. Instead, we expect participants to bring an open-minded attitude and a genuine willingness to engage with the complexities of sustainability. Curiosity, critical reflection, and a readiness to challenge assumptions are essential. These qualities will enable students to work effectively with the course concepts and contribute meaningfully to class discussions and project work.

Learning objectives

- **Integrate Foresight and Sustainability Models:**
  You will learn to apply foresight methods together with core sustainability frameworks—such as the Circular Economy and Butterfly Model—to identify strategic opportunities for sustainable innovation.
- **Understand and Apply Transformational Innovation:**
  You will develop the ability to distinguish between incremental and transformational innovation and assess how organizations can shift toward regenerative and sustainable business practices.
- **Use Human-Centered Design for Sustainability Challenges:**
  You will acquire skills in applying human-centered design to uncover user needs, behaviors, and system dynamics that shape sustainability-oriented innovation.
- **Translate Insights into Strategic, Managerial Action:**
  You will learn to synthesize foresight, sustainability models, and design thinking into actionable strategies and recommendations for organizational decision-making.

Content

Since the publication of the United Nations' 17 Sustainable Development Goals (SDGs) in 2015 (United Nations Statistics Division, 2020), it has become increasingly evident that societies, governments, and organizations must pursue transformative pathways to address urgent global challenges such as poverty, public health, climate change, and equitable access to education. Despite notable progress, the current moment remains critical, demanding accelerated and more systemic responses to these complex issues.

In recent years, economists, business leaders, and policymakers have advanced various strategic approaches to support the SDGs. Popular concepts—including "win-win" strategies, Circular Economy models, and Reverse Innovation—have promised alignment between corporate profitability and sustainability objectives. Yet, scholars increasingly warn that many of these

paradigms risk oversimplifying the inherent trade-offs in sustainable development and are often grounded in flawed mental models (The Dangerous Allure of Win-Win Strategies, 2020). Understanding these limitations is essential for designing innovation strategies that are both realistic and impactful.

Course Focus

This three-day course equips students with the competencies needed to conceptualize and design sustainable innovations using human-centered design methodologies. Throughout the course, students will explore how sustainability transitions can be shaped through rigorous foresight, behavioral insight, and systemic thinking. Key topics include:

- **Foresight and Future Thinking**: Understanding how scenario analysis, trends, and signals of change inform long-term sustainability strategies and innovation opportunities.
- **Sustainability Models and Transformational Innovation**: Applying frameworks such as the Circular Economy and the Butterfly Model to identify leverage points for regenerative and systemic innovation.
- **Design-Driven Innovation**: Integrating behavioral insights, user research, and contextual understanding to support sustainable solution development.
- **Design Thinking for Sustainability**: Applying human-centered design principles to craft, prototype, and evaluate strategies for sustainable transformation within organizations.

Course Format

The course combines multiple learning formats to create a rich, practice-oriented experience:

- **Lectures** to establish core foundations in sustainable innovation, foresight, and design.
- **Guest sessions** with senior practitioners and experts (subject to availability).
- **Case-based work and hands-on exercises** that allow students to apply frameworks and design methods to real-world sustainability challenges.

References

The Dangerous Allure of Win-Win Strategies. (2020, Winter).
United Nations Statistics Division. (2020). *The Sustainable Development Report.*

Structure and indications of the learning and teaching design

- This course counts 3 credits. Accordingly, the total workload for students is 90 hours. This includes self-study, campus time, and all examinations.
- The structure of the contact study is planned as follows: 3 full days of lectures within a block course, 2 hours of introductory lectures, 4 hours of guest lectures (requested), and 5 hours of interactive contact in a blended learning format.
- The course is conducted in presence in St. Gallen and in a blended learning format.

Further explanation of the teaching and learning design:
The program starts on program days at 8.30 am and runs until 16 pm with short breaks and a lunch break. For the casework, it will be required to work together in small groups. We will have a 2-hour preparation meeting at the beginning of the semester.

Literature

Literature

- UN Report: https://unstats.un.org/sdgs/report/2020/
- The Dangerous Allure of Win-Win Strategies. (2020, Winter). https://ssir.org/articles/entry/the_dangerous_allure_of_win_win_strategies

Additional information

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Examination information

Examination sub part/s

1. Examination sub part (1/3)

Examination modalities

Examination type	Presentation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Group work group grade
Weighting	30%
Duration	--

Languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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2. Examination sub part (2/3)

Examination modalities

Examination type	Active participation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Synchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	10%
Duration	--

Languages

Question language: English
Answer language: English

Remark
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Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids
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3. Examination sub part (3/3)

Examination modalities

Examination type	Written work
Responsible for organisation	decentral
Examination form	Written work
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	Off Campus
Grading type	Individual work individual grade
Weighting	60%
Duration	--

Languages
Question language: English
Answer language: English

Remark
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Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids
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Content

The examination will be based on the topics presented and discussed as part of the course. Further, all guest lectures are relevant for the examination too as well as the case study work as part of the lecture program.

Literature
In addition to the literature that is specified in section "course content", the handed-out case studies are relevant as well.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW21) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 13 (Monday, 23rd March 2026) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 04 (Thursday, 22nd January 2026);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 12 (Monday, 16 March 2026);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 14 (Monday, 30 March 2026);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 15 (Monday, 06 April 2026).